02464 Artificial Intelligence and Human Cognition

Week 8&9:

classification basics

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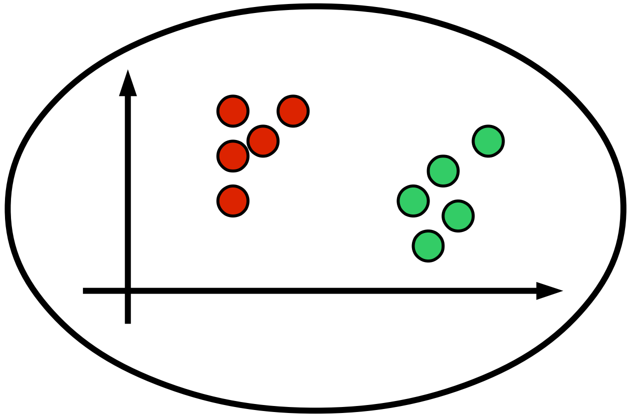
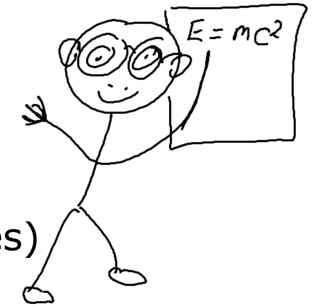
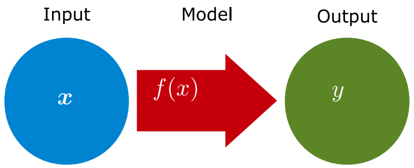
Supervised Learning

Input data $\mathbf{x}_n$ and output y_n



(Generalize from known examples)





- **Data**
  - Inputs and outputs (this is what we are given)

$$\{\boldsymbol{x}_n, y_n\}_{n=1}^N$$

- **Model**
  - Function that maps inputs to outputs (what we are trying to determine)

$$f(\boldsymbol{x})$$

- **Cost function**
  - Dissimilarity measure between observation and prediction (how we tell if a model is good or bad)

$$d\big(y, f(\boldsymbol{x})\big)$$

- **Types of supervised learning**
  - Regression: continuous output y
  - Classification: discrete output y

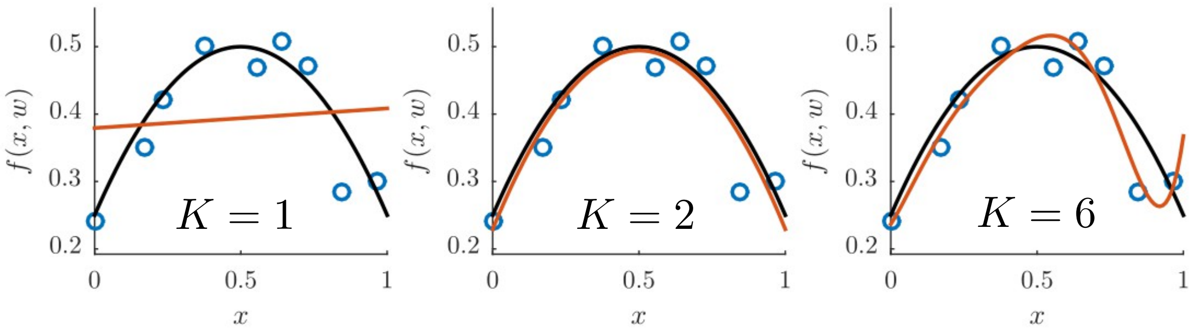
Classification

- **definition:** learning a function that maps a data object to a discrete class

- **why classify?**
  - descriptive modeling
    - explain/understand the relation between features and class
    - e.g., insight into neural mechanisms that distinguish two conditions
  - predictive modeling
    - predict the class of a new data object

Assess model performance: Training error

- (regression example on non-linearly transformed inputs: polynomials)
- suppose you train 3 models on a dataset of 9 observations

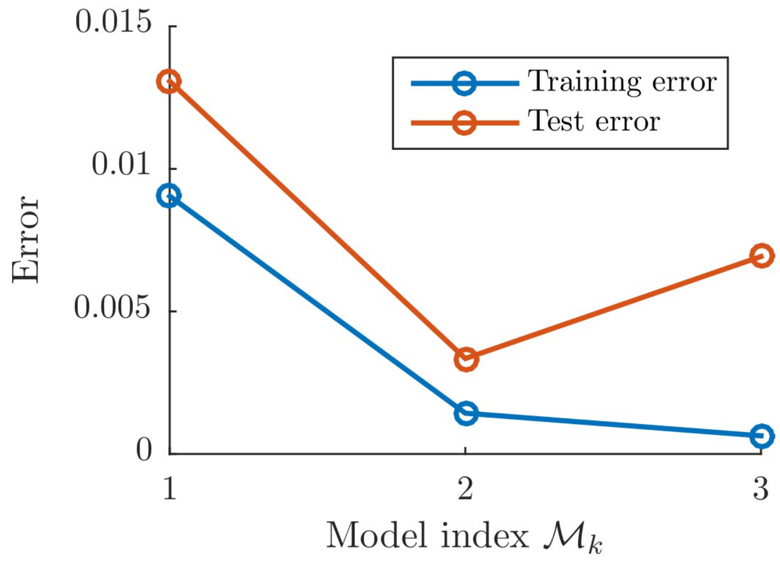


$$E^{\text{train}}_{\mathcal{M}_k} = \frac{1}{N^{\text{train}}} \sum_{i \in \mathcal{D}^{\text{train}}} (y_i - f_{\mathcal{M}_k}(x_i, \boldsymbol{w}))^2$$

Never estimate how well the model performs by its predictions on data it was trained on!

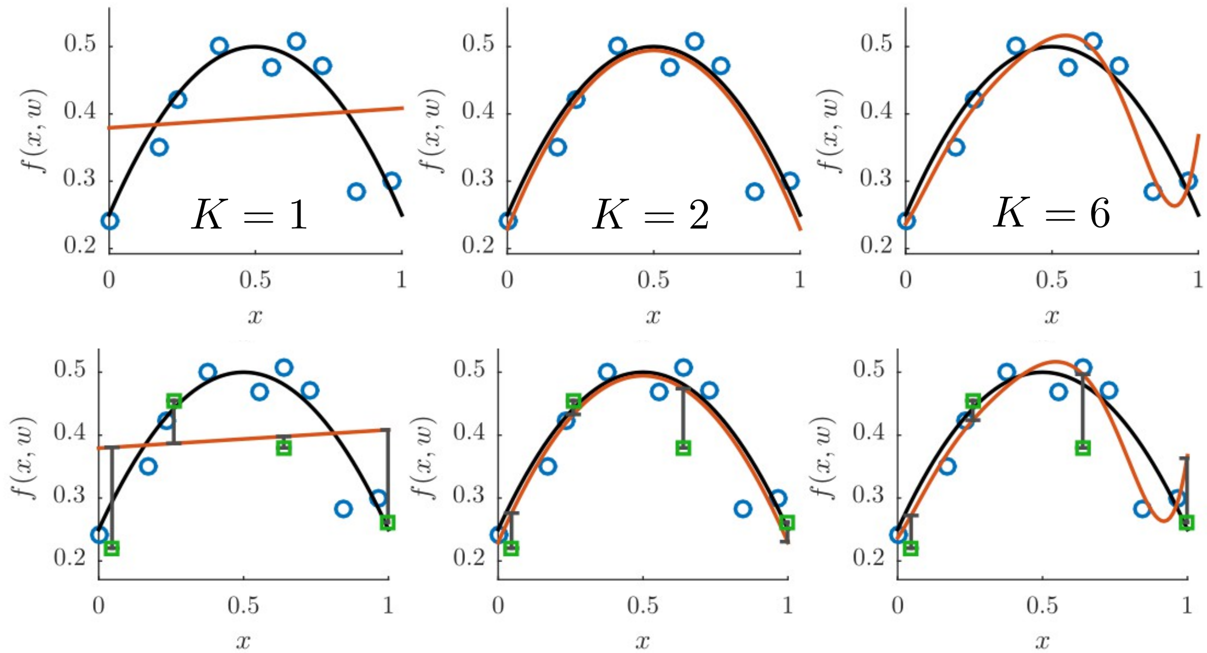
Overfitting

- **Overfitting**: the training error usually decreases for overly complex models while the test error increases
- Test error is the more "true" error



Assess model performance: Test error

- test error is obtained by testing the trained models on new data



$$E^{\text{train}}_{\mathcal{M}_k} = \frac{1}{N^{\text{train}}} \sum_{i \in \mathcal{D}^{\text{train}}} (y_i - f_{\mathcal{M}_k}(x_i, \boldsymbol{w}))^2$$

$$E^{\text{test}}_{\mathcal{M}_k} = \frac{1}{N^{\text{test}}} \sum_{i \in \mathcal{D}^{\text{test}}} (y_i - f_{\mathcal{M}_k}(x_i, \boldsymbol{w}))^2$$

How do we divide our dataset into training and test data?

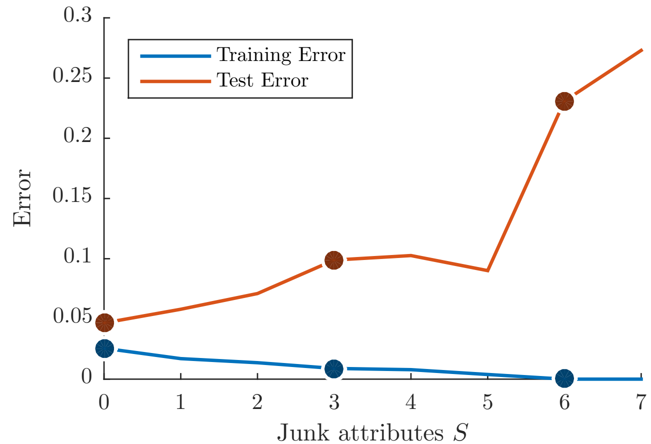
- And how to measure the generalization error? i.e., how our model performs on average, assuming we had an infinite amount of data to test it on?
- K-fold cross-validation
  - Divide the sample data into K parts, each containing N/K observations
  - Use K-1 of the parts for training, and 1 for testing
  - Repeat K times, rotating the test set
  - Choose K = 10

$$E_{\mathcal{M}}^{\text{gen}} \approx \sum_{k=1}^{K} \frac{N_k^{\text{test}}}{N} E_{\mathcal{M},k}^{\text{test}},$$

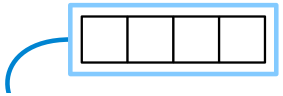
  - Advantage? Each data point is used once in the test set

Sequential feature selection

- Suppose you want to do a linear regression to predict a patient's survival rate after an operation
- You have 3 features which correspond to age, room number of the patient, and length of hospital stay, respectively
- Only the first and last feature are relevant, so could just consider the data set without the second feature
- The more "junk" features you have, the more the model will overfit the training set
- If we consider all the possible feature sets, this will be too costly!
- Sequential feature selection only considers a subset of models

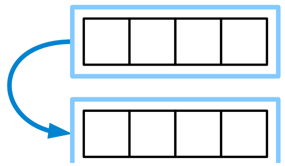


Sequential forward feature selection



- Start with no features

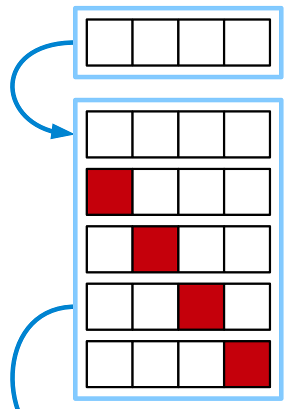
Sequential forward feature selection



- Start with no features
- Compute cross-validation error for
  - Current feature subset

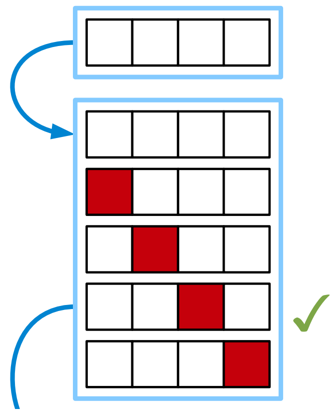
Sequential forward feature selection

- Start with no features
- Compute cross-validation error for
  - Current feature subset
  - All subsets equal to the current
    + one added feature



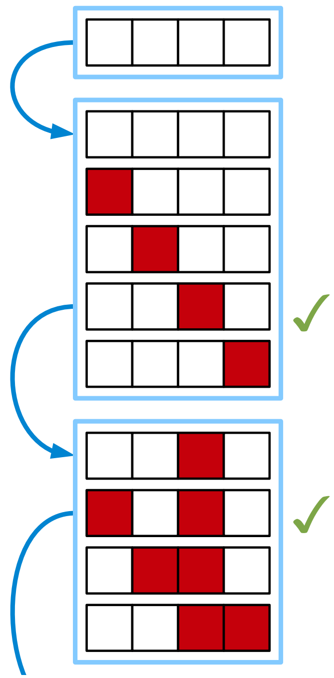
Sequential forward feature selection

- Start with no features
- Compute cross-validation error for
  - Current feature subset
  - All subsets equal to the current + one added feature
- Choose best subset



Sequential forward feature selection

- Start with no features
- Compute cross-validation error for
  - Current feature subset
  - All subsets equal to the current
    + one added feature
- Choose best subset
- Repeat until no further improvement



Sequential forward feature selection

- Start with no features
- Compute cross-validation error for
  - Current feature subset
  - All subsets equal to the current + one added feature
- Choose best subset
- Repeat until no further improvement
- Feature set tells us which are the best predictors